POSTER: GEM-PLI SIDE CHANNEL IN GPON INFERRING ENCRYPTED STREAMING TRAFFIC

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INTRODUCTION & MOTIVATION

1.As fiber internet access has become dominant in Network Infrastructures

ssive optical network (PON) technologies-accounted for 54.2% of Fiber To The Home (FTTH) market in 2024

it is important to ensure traffic confidentiality.

2.GPON supports AES-based downstream traffic encryption on GEM Layer.

3.However, some fields of GEM Header remain plaintext

4.recent studies have demonstrated that such metadata like PLI can be used to infer victim's activity.

5.we researched whether the plaintext PLI field in GPON GEM header can be used to leak victim's privacy on the same optical splitter

BACKGROUND

GPON =

METHODOLOGY

• 1. Premise The attacker has an optical capture device (e.g., FPGA with optical TAP, modified ONU that can receive raw GTC frames) connected to splitter and passively records downstream from OLT. • * The attacker does not need to register their ONU/ONT on OLT; so even if they use receive-only optical tap, they can entirely record downstream records without transmitting. • 2. Collect GEMs Then, attacker extracts GTC frames from downstream traffic and parses GEM frames. • 3. Convert GEM PLI to dataset Then, the attacker filters GEM frames for a victim's Port-ID and records PLI values with timestamps. • 4. Dataset format The dataset must include start & end time and duration and PLI and sequence information. • 5. Classification via database matching The attacker matches the captured PLI time series to a pre-collected database of PLI traces (built from known videos) and selects the closest match to infer the victim's video.

EXPERIMENT

we developed a software GPON downstream simulator that reproduces the process of receiving GTC frames, extracting GEM frames, and exporting plaintext GEM header fields into datasets. At first we played each of 10 videos three times in our linux based system to produce packet capture datas (PCAPs) where other streaming traffic(e.g., VNC) is restricted to minimize interference. then we simulated that PCAPs packet capture file to records the resulting PLI values and timestamps into CSV traces

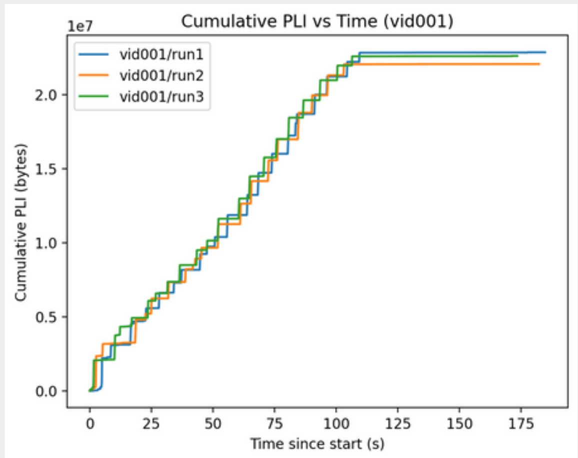


fig. Cumulative PLI - Time Grapgh for one video showing that greatly identical slopes between three runs, PLI - Time Grapgh of others is publicly available on our github repository

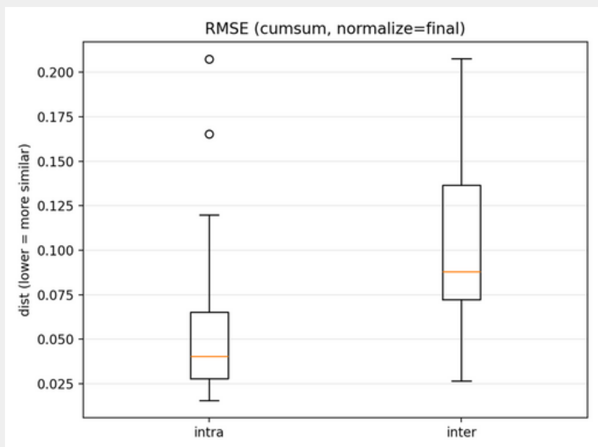
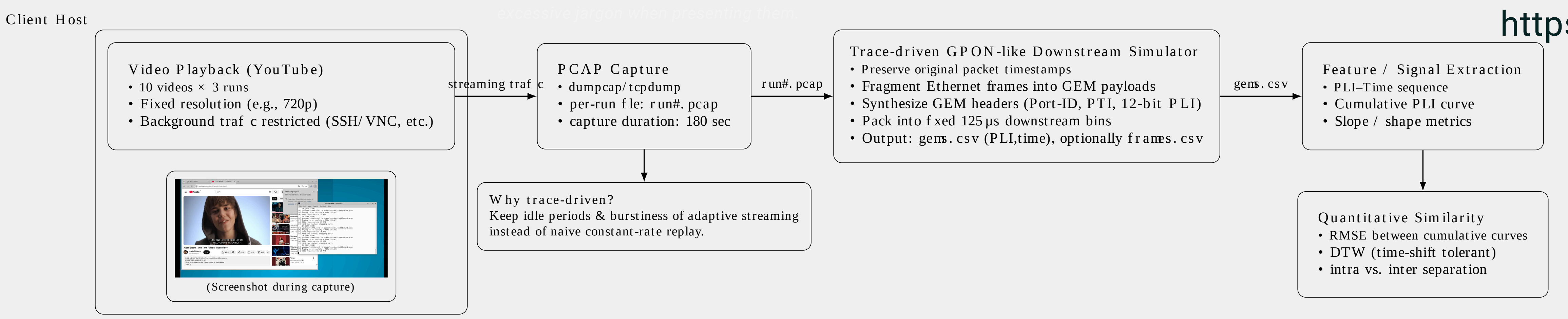


fig.RMSE of Cumulative PLI
left: Same Video(Intra)
Right: Different Video(Inter)
**Lower=More Similar*
(AUC:0.813)



End-to-end experimental pipeline: YouTube playback and PCAP capture (180 s), trace-driven GPON-like downstream simulation producing GEM-PLI sequences, followed by quantitative similarity analysis.

CONCLUSION

GPON Network may be vulnerable to streaming inferring attack.
Payload encryption alone does not ensure confidentiality of traffic.
RMSE: 0.040 (Same Video) / 0.088 (Different Video)
(AUC:0.813)

FUTURE WORK

Future Work will be conducted in experimental testbed with Real OLT and Optical Splitter. We will use FPGA to collect GEM frames.
when adopted CNN, it is possible to distinguish video Ids using captured PLI values.

DATASET & CODES

Our datasets and codes are publicly available on our github
https://github.com/hjaka08/GPON

REFERENCES

[1] ITU-T Recommendation G.984.3, Gigabit-capable passive optical networks (GPON): Transmission convergence layer specification.

[2] S. Bae KAIST, "Watching the Watchers: Practical Video Identification Attacks in LTE Networks," in USENIX Security, 2022.

[3] Cisco, "Understand GPON Technology," Troubleshooting TechNotes.

[4] github repository - https://github.com/hjaka08/GPON

[5] Fiber-to-the-Home (FTTH) Market Size & Share Analysis - Growth Trends and Forecast (2025 - 2030)

[6] Traffic spills the beans- A robust video identification attack against YouTube - Elsevier Computers & Security (2024)

[7] Breaking Through the Diversity: Encrypted Video Identification Attack Based on QUIC Features - ESORICS 2024